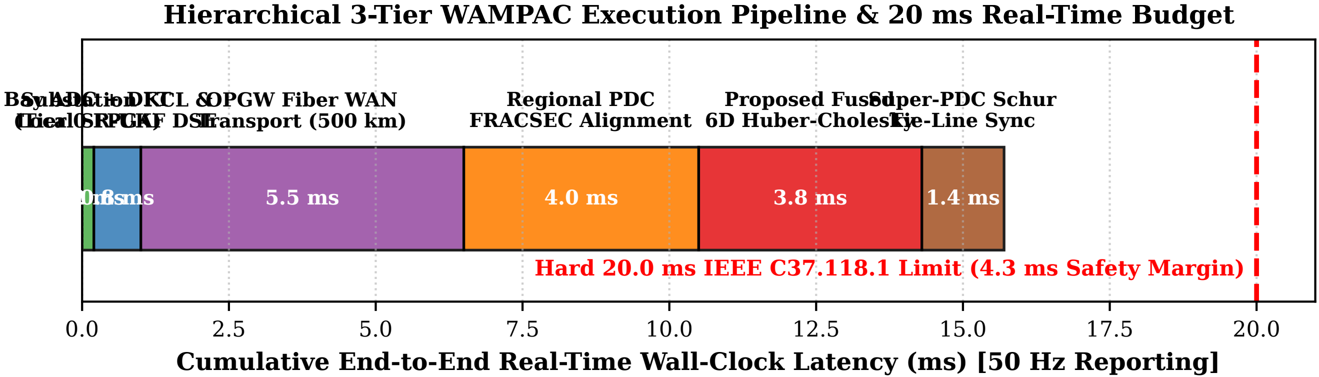Hierarchical 3-Tier WAMPAC Execution Pipeline & 20 ms Real-Time Budget
Bay ADC + DFT
(Tier 0 PMU)
Substation KCL &
Local SRUKF DSE
OPGW Fiber WAN
Transport (500 km)
Regional PDC
FRACSEC Alignment
Proposed Fused
6D Huber-Cholesky
Super-PDC Schur
Tie-Line Sync
0.2 ms
0.8 ms
5.5 ms
4.0 ms
3.8 ms
1.4 ms
Hard 20.0 ms IEEE C37.118.1 Limit (4.3 ms Safety Margin)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Cumulative End-to-End Real-Time Wall-Clock Latency (ms) [50 Hz Reporting]